

Rotations

1. Euler Angle

1. Euler Angle is very intuitive

2. Euler Angle to Rotation matrix

1. Rotation about principal axis is represented as:

$$2. R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}$$

$$3. R_y(\beta) = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}$$

$$4. R_z(\gamma) = \begin{bmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

5. $R = R_z(\alpha)R_y(\beta)R_x(\gamma)$ for arbitrary rotation

3. Inspection from Learning perspective

1. Euler Angle is not unique for some rotations. For example $R_z(45^\circ)R_y(90^\circ)R_x(45^\circ) = R_z(90^\circ)R_y(90^\circ)R_x(90^\circ)$

$$2. = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

3. Gimbal lock:

4. Df is rank-deficient at some $\theta \Rightarrow$ some movement in $T_{f(\theta)}(SO(3))$ cannot be achieved.

5. For example, when $\beta = \pi/2$, $R = R_z(\alpha)R_y(\pi/2)R_x(\gamma)$

$$6. = \begin{bmatrix} 0 & 0 & 1 \\ \sin(\alpha + \gamma) & \cos(\alpha + \gamma) & 0 \\ -\cos(\alpha + \gamma) & \sin(\alpha + \gamma) & 0 \end{bmatrix}$$

7. since changing α and γ has the same effects, a degree of freedom disappears!

	Inverse?	Composing?	Any local movement in $SO(3)$ can be achieved by local movement in the domain?
Rotation Matrix	✓	✓	N/A
Euler Angle	Complicated	Complicated	No
Angle-axis	✓	Complicated	?
Skew-symmetrical Matrix	✓	Complicated	?
Quaternion	✓	✓	✓

? means no singularity with single exceptions

References:

1. ML meets geometry, Hao Su Lab, UCSD, 2022.